

<i>Please ensure this tracking sheet is completed as a group, as it will be used to monitor your individual participation within the team.</i>						
Team Members	Role/ Responsibility	Email				
Milliam Mukamukiza	Data preparation, data cleaning, missing values, non-numeric data, explained variance, dynamic PC selection, documentation, and create before/after PCA visualizations	m.mukamukiz@alustudent.com				
Gasana	Covariance, eigenvalues/eigenvectors, performance optimization, and benchmarking	g.irakora@alustudent.com				
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Task Allocation and Tracker						
	Task Allocated	Assigned Member	Deadline	Completion Status (Not Started/In Progress/Completed/Did not participate)	Reviewed by Team? (Yes/No)	Comments
	ind and select suitable African dataset with ≥ 7 columns, missing values and non-numeric data	Milliam	28 Sep 2026	Completed	Yes	
	Inspect dataset and identify missing values and non-numeric columns	Milliam	28 Sep 2026	Completed	Yes	
	Clean missing values and encode non-numeric data using NumPy	Milliam	28 Sep 2026	Completed	Yes	
	Implement data centering and covariance matrix calculation	Gasana	28 Sep 2026	Completed	Yes	
	Calculate and sort eigenvalues/eigenvectors	Gasana	28 Sep 2026	Completed	Yes	
	Project centered data onto principal components	Gasana	28 Sep 2026	Completed	Yes	

	Calculate explained variance and dynamically select PCs	Milliam	28 Sep 2026	Completed	Yes	
	Create before/after PCA visualizations	Milliam	28 Sep 2026	Completed	Yes	
	Optimize, benchmark, review notebook and prepare final submission	Gasana	28 Sep 2026	Completed	Yes	

Meeting Attendance Log

Meeting Date	Agenda	Facilitator	Attendees	Key Discussion Points	Next Steps	
25 Sep 2026	Review of assignment questions and understanding the requirements	Gasana	Gasana and Milliam	We went through the assignment questions together and discussed what each question required. We helped each other understand the PCA tasks and shared our initial ideas on how to approach the work.	Agree on the approach for completing the questions and divide the work.	
26 Sep 2026	Discussion of approaches and sharing ideas	Milliam	Gasana and Milliam	We discussed our ideas on how to complete the PCA assignment and shared our understanding of the different tasks. We compared our approaches and clarified areas that were not clear.	Finalize task responsibilities and begin working on the assigned sections.	
27 Sep 2026	Task allocation and work planning	Gasana	Gasana and Milliam	We discussed how the different tasks would be completed and divided the work between us. We agreed on the responsibilities of each member and how we would support each other during the process.	Complete the assigned tasks and communicate progress through WhatsApp.	
28 Sep 2026	Completion of assigned PCA tasks	Milliam	Gasana and Milliam	We worked on the assigned tasks, including data preparation, PCA calculations, explained variance, visualization, and performance-related tasks. We also communicated through WhatsApp to update each other on our progress.	Review each other's work and identify errors or areas that need correction.	

29 Sep 2026	Review, error checking, and consolidation of work	Gasana	Gasana and Milliam	e came together to show and review what each person had completed. We compared the results, identified some errors, discussed corrections, and brought the separate work together for the final submission. We also continued communicating through WhatsApp.	Correct the identified errors, finalize the notebook and contribution sheet, and prepare the final submission.	
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